Course Title: **LINEAR ALGEBRA**
Instructors: **Aeyaz J Kayani, Sajal S Rana**
Exam: **Midterm**

Class ID: **MATH 205**
Total Marks**: 60**
Duration**: 75 minutes**

	Marks Obtained
Q. 1	
Q. 2	
Q. 3	
Q. 4	
Q. 5	
Q. 6	
Q. 7	
Total	

Instructions for Students:

a. Read the following instructions carefully.
b. Please ensure that your mobile phone is switched off.
c. **All your personal belongings must be placed at the designated area of the examination hall.**
d. Write your Name, ID, Subject, Section, Instructor's name, and date in the space provided below.
e. Remember to sign the exam attendance sheet circulated by the head invigilator / proctor.
f. You may use half of an A4 single-sided handwritten formula sheet, which must be submitted with the rest of your work at the end of the exam.
g. Answer **all** questions in the exam.
h. You must provide complete working and reasoning for all steps of your solution.
i. If you need to leave the room for any reason, please raise your hand and take permission from head invigilator / proctor.
j. After finishing your exam, raise your hand and give your answer book to the head invigilator / proctor.

Name: _____

Student ID: _____ Section: _____

Subject: _____

Instructor: _____

Date: _____

ROUGH WORK

Question 1: [10 Marks]

Show that if A has a row of zeros and B is any matrix for which AB is defined, then AB also has a row of zeros.

Question 2: [5 Marks] Show that the most general matrix that commutes with $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ is of the form $\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$.

Question 3: [5 Marks]

Let A be a square matrix, such that $A^{r+s} = A^r A^s$. Is this valid for negative integers r, s ?

Question 4: [10 Marks] Consider an $n \times n$ system of linear equations $A\,\boldsymbol{x} = B$, where $\boldsymbol{x}$ is a column vector consisting of variables $x_1, x_2, \ldots, x_n$. Let D_k be the matrix obtained when the k-th column of I_n is replaced by $\boldsymbol{x}$. Let $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_n$ denote the columns of I_n ; and $\boldsymbol{c}_1, \ldots, \boldsymbol{c}_n$ denote the columns of A. You can also assume that we know that $\boldsymbol{e}_k = A^{-1}\boldsymbol{c}_k$ for all $1 \leq k \leq n$. If A_k is the same matrix as in the proof of Cramer's Rule that was shown in class (i.e. obtained by removing the k-th column of A with $\boldsymbol{x}$, then show that $\det(D_k) = \frac{\det(A_k)}{\det(A)}$.

Question 5: [10 Marks] Consider the set V of all positive real numbers, and define the operations on V as follows:

- Vector addition: $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{uv}$
- Scalar multiplication: $k\ \boldsymbol{u} = \boldsymbol{u}^k$

Prove that V is a vector space.

Question 6: [10 Marks]

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \dots, \boldsymbol{v}_r\}$ be a set of vectors in the Euclidean space $\mathbb{R}^n$. Show that if $r > n$, then S is not linearly independent.

Question 7: [10 Marks]

Prove that for a given square matrix A, A is non-singular if and only if the reduced row echelon form of A is the identity matrix of the same order as A.

ROUGH WORK